MIL-A-48027A (PA)
30 June 1975
Superseding
MIL-A-48027 (MU)
14 January 1972

MILITARY SPECIFICATION

ADAPTER-BOOSTER, BOMB, NOSE, THERMALLY PROTECTED, M148E1, LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one type of booster designated as Adapter-Booster, Bomb, Nose, M148E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals form a part of this specification to the extent specified herein.

*SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for.
MIL-P-46464 - Pellets, Tetryl.

*STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

*PUBLICATIONS

ASTM-D1692 - Flammability of Plastic Sheeting and Cellular Plastics.

FSC: 1325

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ASTM-D2044 - Moisture Content in Paper and Paperboard by Toluene Distillation.
ASTM-D792-66-Specific Gravity and Density of Plastics by Displacement

*DRAWINGS

U.S. ARMY PICATINNY ARSENAL

9270711 - Box, Packing, Ammunition for Adapter Boosters M147, M148, M148E1, M149 or XM150
9270710 - Support, Packing, Ammunition for Adapter Booster M148 or Adapter Booster M148E1
9272263 - Adapter-Booster, Bomb, Nose, Thermally Protected, M148E1, Loading Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*For meaning of asterisk see 6.7.

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

*3.2 Adapter-Booster assemblies. - The assemblies shall comply with all requirements specified on Drawing (dwg.) 9272263 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Sealing of barrier bag. - When the barrier bag is required, all included air shall be removed prior to sealing.

*3.4 Packing supports. - The packing supports when made of molded polystyrene foam, shall not contain moisture in excess of 0.4 percent, shall not exhibit an electrical potential in excess of 600 volts when rubbed with a rayon or silk cloth, and shall be self extinguishing (flammability) in accordance with Standard ASTM-D1692.

3.5 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity,

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warpeage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

3.6 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Pellet, Booster	8799590	10
Support, Packing	9270710	15
Booster Loading Assembly	9272263	10

*4.3.2 Inspections to be performed. - See MIL-A-48078 and paras 4.4.2.1 thru 4.4.2.7, testing per para. 4.4.3, and Table I specified herein.

*4.3.3 Rejection. - (See MIL-A-48078)

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*TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-A-48027A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Adapter Booster Assembly and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Pellet, Booster (Dwg. 8799590)				
	Examination for Defects	10		3.2	4.4.2.7
	Composition test	10		3.2	MIL-P-46464
	Specific Gravity	10		3.2	4.5.4
	Support, Packing (Dwg. 9270710)				
	Moisture Content test	5		3.4	4.5.1
	Static Potential Charge test	5		3.4	4.5.2
	Flammability test	5		3.4	4.5.3
	Booster Loading Assembly (Dwg. 9272263)				
	Examination for Defects	10		3.2	4.4.2.4

NOTES:

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4.4 Quality conformance inspection

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

*4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48027A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.1	Loading Assembly Prior to Assembly of Closing Cup				9272263 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Pellet missing		0.40%	3.2	Visual
Major B					
131	Base disc, disc or spacer missing		0.40%	3.2	Visual
132	Cement missing from closing cup area (360 degrees)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-A-48027A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.2	Loading assembly, prior to assembly of sleeve and adapter				9272263 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Distance end to closing cup, minimum (min.). (3 places - equally spaced)		0.40%	3.2	Gage
132	Insulating disc missing		0.40%	3.2	Visual
133	Booster well seal damaged		0.40%	3.2	Visual
134	Cement missing from hole in closing cup		0.40%	3.2	Visual
Minor					
701	Evidence of poor workmanship		1.00%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48027A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272263
*4.4.2.3	Loading Assembly Prior to Closing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Concentricity of casing with external thread		0.40%	3.2	Gage
132	Cement on thread not visible (360 degrees), when applicable		0.40%	3.2	Visual
133	Sleeve not seated (either end)		0.40%	3.2	Visual/Manual
134	Insulating washer missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48027A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272263
#4.4.2.4	Loading Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Large external thread damaged		0.40%	3.2	Visual
Major B					
131	Closing plug missing		0.40%	3.2	Visual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Coating missing from marking or damaged area		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.5	Box, Packing, Prior to Closing				9270711
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Desiccant bag missing		0.40%	3.2	Visual
132	Contents loose		0.40%	3.2	Visual/Manual
133	Support damaged or missing		0.40%	3.2	Visual
134	Barrier bag improperly sealed		0.40%	3.2	Visual
135	Included air not removed from barrier bag		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-48027A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.6	Box, Packing, Closed				9270711
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Contents loose		0.40%	3.2	Visual
132	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
133	Board broken		0.40%	3.2	Visual
134	FSN number missing or misleading		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Strapping improperly positioned		0.65%	3.5	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.7	Pellet, Booster				8799590
					NEXT HIGHER ASSEMBLY 9272263
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Outside diameter		0.40%	3.2	Gage
132	Inside diameter		0.40%	3.2	Gage
133	Length		0.40%	3.2	Gage
134	Cracks or laminations		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Booster pellet (see dwg. 8799590) - Major B defect. - The booster pellet shall be inspected and tested in accordance with Specification MIL-P-46464, Rejection criteria shall be as specified in that specification.

*4.4.3.2 Packing supports (see 3.4) - Major B defects; Moisture content, Static potential charge, and Flammability. - A sample of thirty (30) supports shall be selected from each shipment received by the loading plant. One (1) support shall be selected from the approximate center of each of 30 individual shipping cartons to make up the required sample. To prevent the loss of moisture from the support, all of the individual supports in the sample shall be sealed in individual polyethylene bags immediately after selection so as to preclude an exchange of occluded gases and the included air kept to a minimum. Ten (10) of the supports shall be subjected to the moisture content determination specified in 4.3.1. The shipment shall be rejected and further testing stopped if one or more supports exhibit excessive moisture content (excess moisture will have an affect on the succeeding tests, therefore performing these tests will serve no useful purpose). If the moisture content is found to be acceptable, the remaining 20 supports shall be tested for static potential as specified in 4.5.2. Two (2) supports from these 20 shall be subjected to the flammability determination specified in 4.5.3. The shipment shall be rejected if two or more supports exhibit excessive static charge, or if one or both flammability samples burn beyond the gage mark described in the ASTM-D1692.

*4.4.3.3 Specific Gravity - Booster Pellet (see dwg. 8779590) - Major B Defect. - The sampling plan shall be in accordance with Table I.

TABLE I

Sample	Sample Size	Cumulative Sample Size	Acceptance Number	Rejection Number
First	10	10	0	2
Retest	10	20	1	2

*4.4.4 Inspection Equipment. - The inspection equipment required to perform the inspections and tests prescribed in

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this specification is identified in the 'Paragraph Reference/ Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.3 herein.

4.5 Test methods and procedures

*4.5.1 Moisture content of packing support

*4.5.1.1 Preparation of specimen. - The total weight of the specimen shall weigh approximately one hundred (100) grams. The specimen shall consist of pieces or sections that are removed sequentially from the sample to yield the maximum specimen weight and size that the test flask can accommodate. The cutting of the support to obtain the specimen shall not be accomplished by use of heat such as hot wires, heated knives, etc. A knife with a serrated edge has been found to be adequate for this purpose. The specimen shall be weighed to the nearest 0.1 gram.

4.5.1.2 Test procedure. - In general, the precautions, calibration, and techniques apparatus prescribed in ASTM D2044 shall be observed. Place 1,200 milliliters of dry (suitably distilled) toluene into the test flask. Add the specimen to the flask and assemble the cover as soon as the specimen is in place. A maximum of five (5) minutes is allowed from the time when the insert is removed from the polyethylene bag to the time when the cover is assembled to the test flask. If this time period is exceeded, discard the contents of the test flask, the test shall not be completed, and a replacement insert and a specimen shall be selected and prepared, adhering to all of the requirements for selecting and preparing the original sample and specimen. The toluene solution containing water from the specimen shall be refluxed by the application of heat to the test flask until the water level in the distillation trap maintains a constant level for 15 minutes minimum. The system shall then be allowed to cool to ambient temperature. Then, 40 to 50 milliliters of toluene, in 10 milliliter increments, shall be poured down the condenser to rinse any water droplets from the surface. The distillation trap shall then be removed from the system, its contents stirred to assure complete separation of the toluene and water. The volume of the water shall be observed and recorded

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as V, estimated to the nearest 0.05 milliliter. The total percentage of moisture content shall be calculated as follows:

$$\text{Percentage of moisture content} = \frac{100\ V}{W}$$

where:

V = Volume of water, in milliliters.
W = Weight of the specimen, in grams.

*4.5.2 Static potential charge. - The support shall be removed from the polyethylene bag, placed on an ungrounded surface, in an environment of not more than 55 percent reliative humidity, and briskly rubbed with a rayon or silk cloth, not less than 25 strokes for a distance of 8 inches minimum, each stroke superimposed upon the preceeding stroke. The cloth used in this test shall be of a size and so arranged that there is no possibility of the operator's hand touching the support during the test. The support shall not be touched in any manner during this induction period or during the measurement which shall follow immediately after the last stroke of holding the detecting head of a Keithley Static Meter No. 600B, with detecting head No. 2501 (or substantial equals) not more than 3/8 inch from the approximate center of the support surface where the rubbing was accomplished.

NOTE: Static potential charge measurements require stringent adherence to manufacturers recommended procedures, including calibration and heat standoff distances. Special attention should be given to preventing disturbance of the coaxial cable between the detector head and the electrometer during test when using the Keithley equipment.

*4.5.3 Flammability. - One specimen shall be prepared as prescribed in ASTM D1692 from each of the supports on the sample and tested as specified therein. The specimen size, conditioning and testing shall be as prescribed in ASTM D1692.

*4.5.4 Specific Gravity - Booster Pellet. - The specific gravity shall be determined in accordance with ASTM Method D-792 Method A-1.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

*5.1.1 Level A. - Preservation and packing shall be as specified on dwg. 9270711.

6. NOTES

*6.1 Intended use. - The Adapter-booster covered by this specification is intended for use in the MK80 series bombs.

*6.2 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

*6.3 Submission of Inspection Equipment Designs for Approval. (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

*6.4 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801.

Asterisks -

6.7 The margins of this specification are marked with as asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1325-A-463

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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